

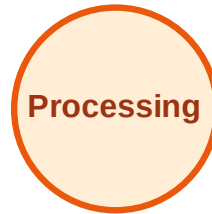
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing

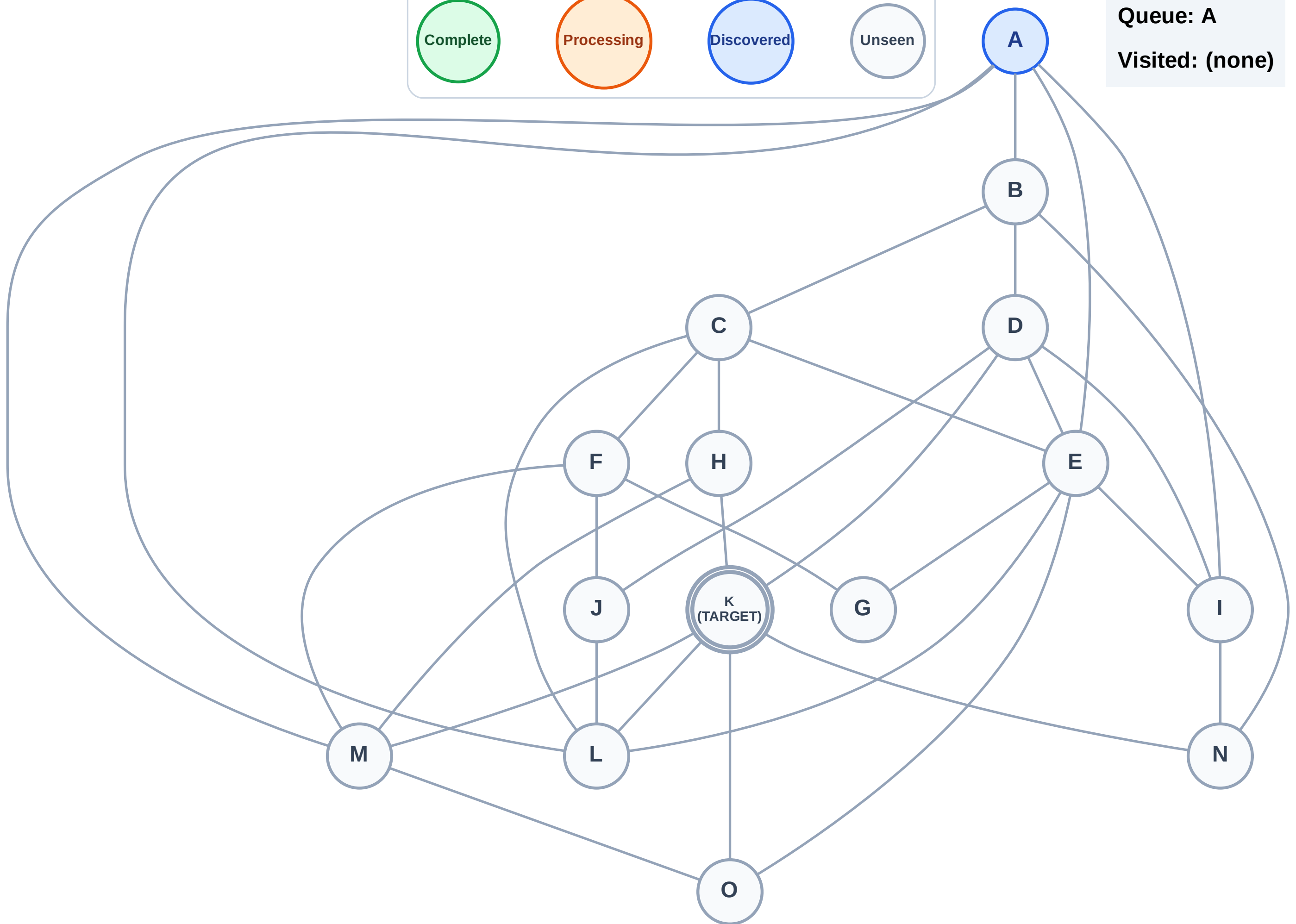


Discovered



Unseen

Queue: A
Visited: (none)



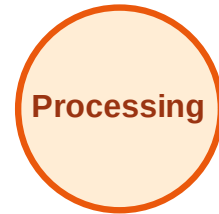
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



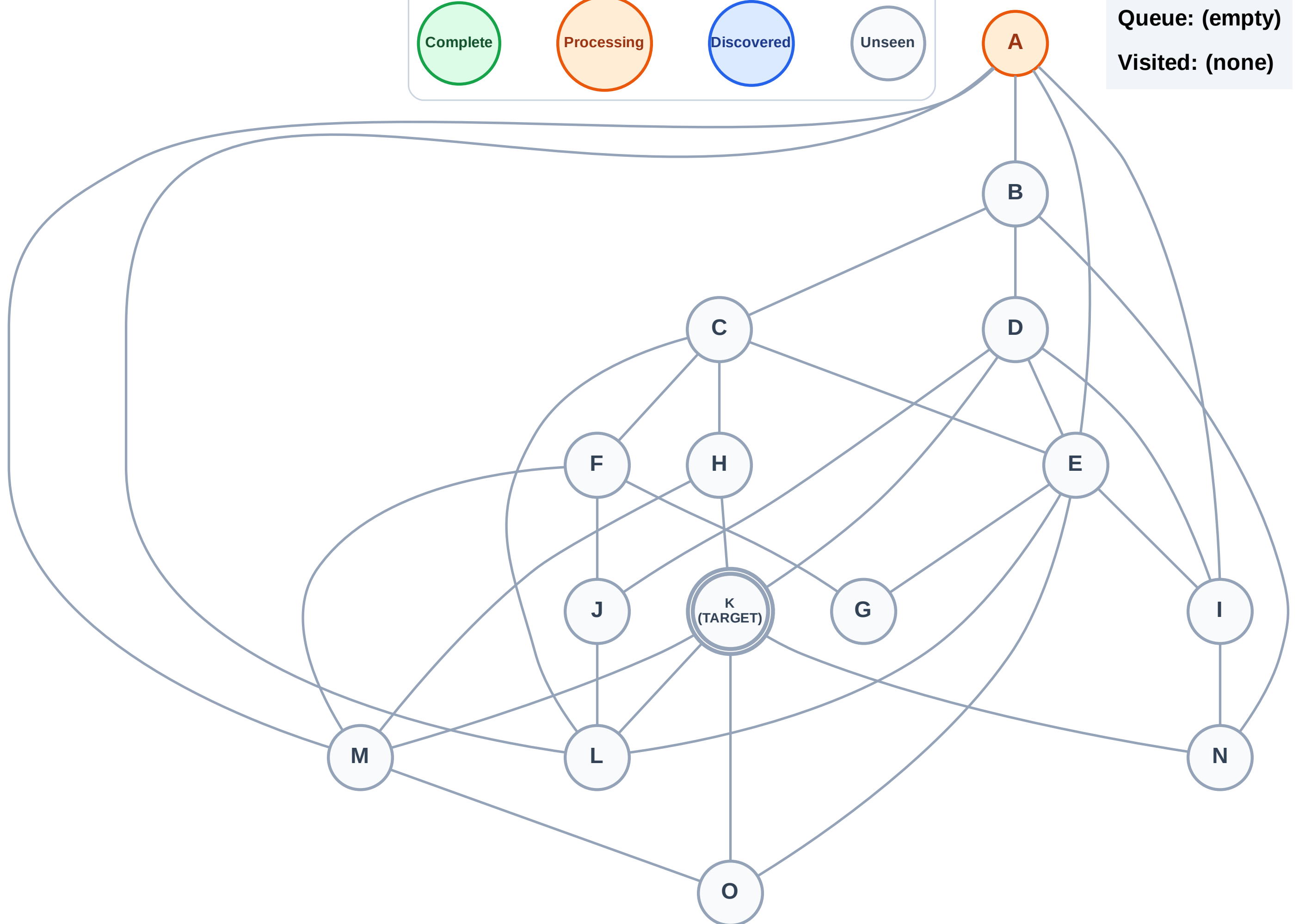
Discovered



Unseen

Queue: (empty)

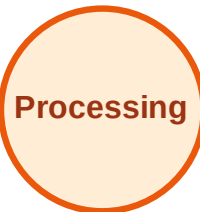
Visited: (none)



BFS Traversal

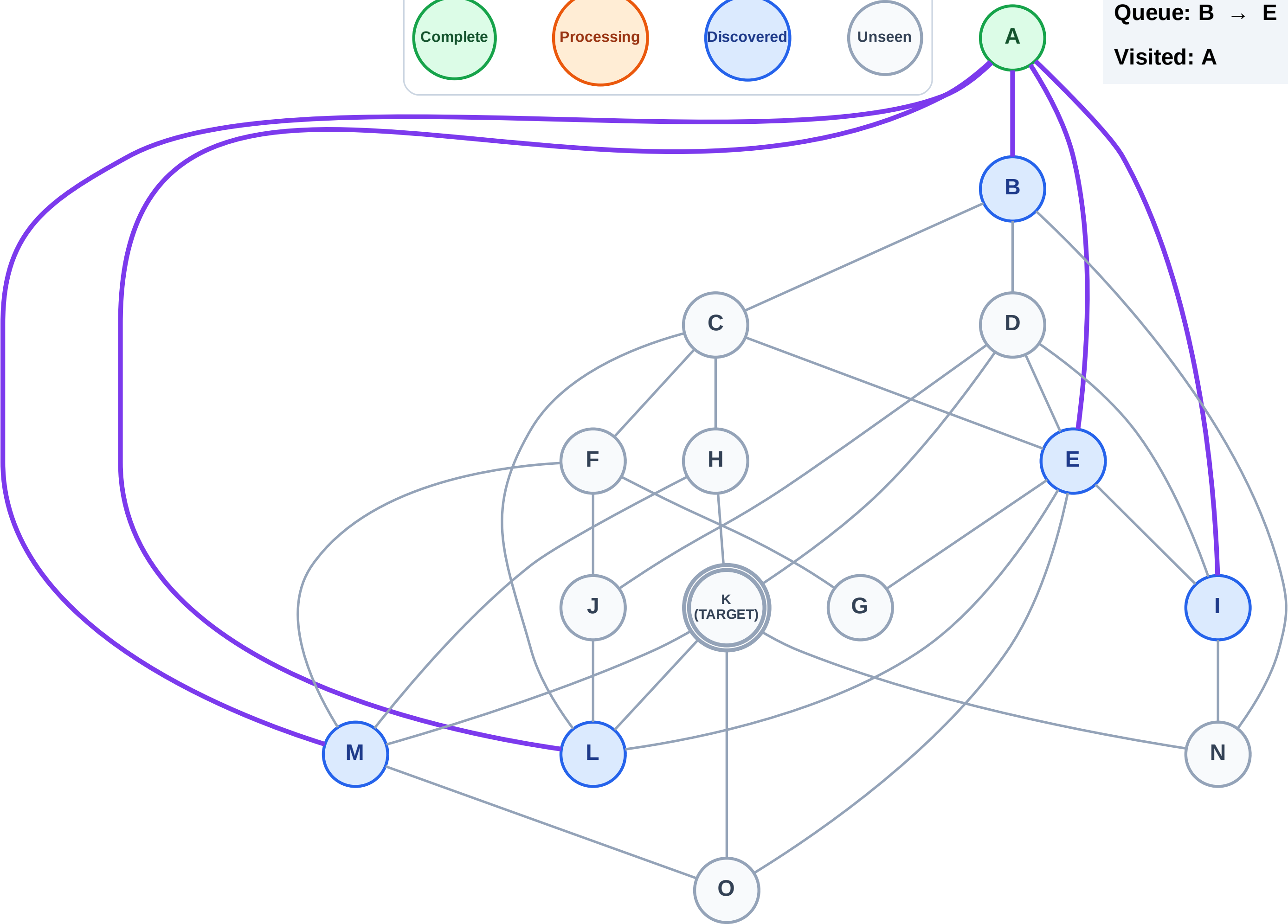
Step 3: Discover B, E, I, L, M from A

Legend



Queue: B → E → I → L → M

Visited: A



BFS Traversal

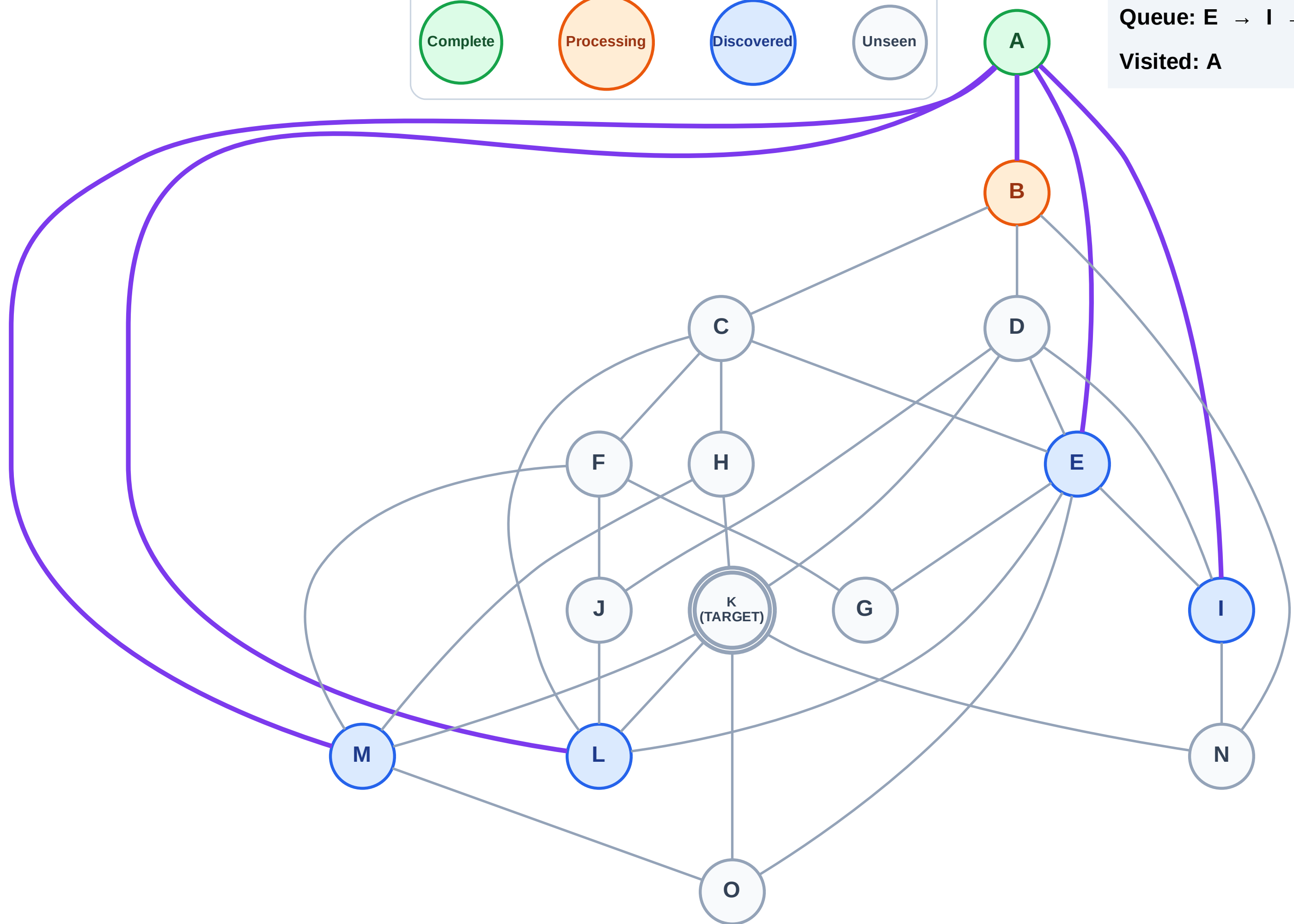
Step 4: Process node B

Legend



Queue: E → I → L → M

Visited: A



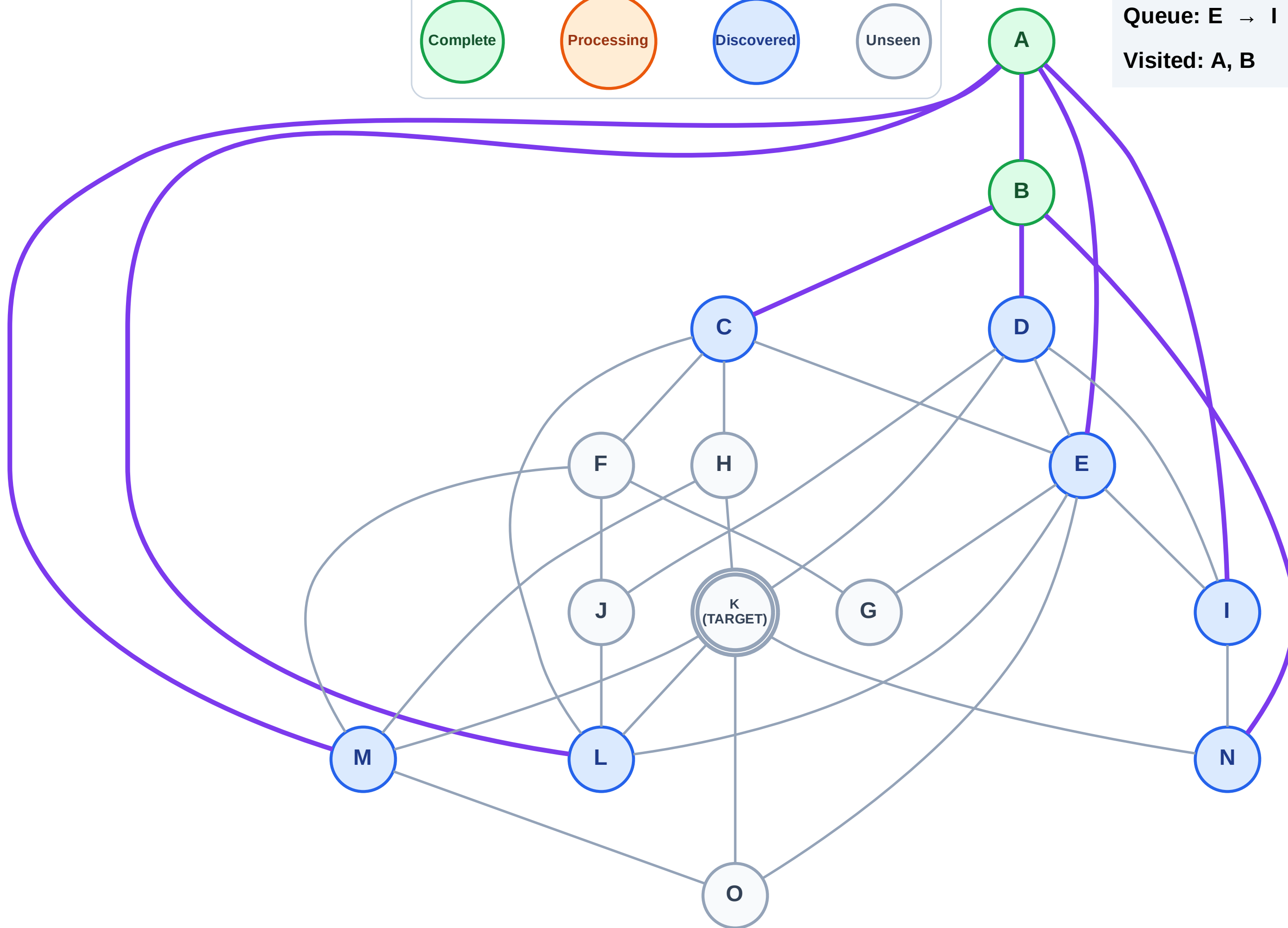
BFS Traversal

Step 5: Discover C, D, N from B

Legend



Queue: E → I → L → M → C → D → N
Visited: A, B



Step 6: Process node E

Legend

The diagram illustrates the four stages of the data science process, each represented by a colored circle:

- Complete** (Green circle)
- Processing** (Orange circle)
- Discovered** (Blue circle)
- Unseen** (Grey circle)

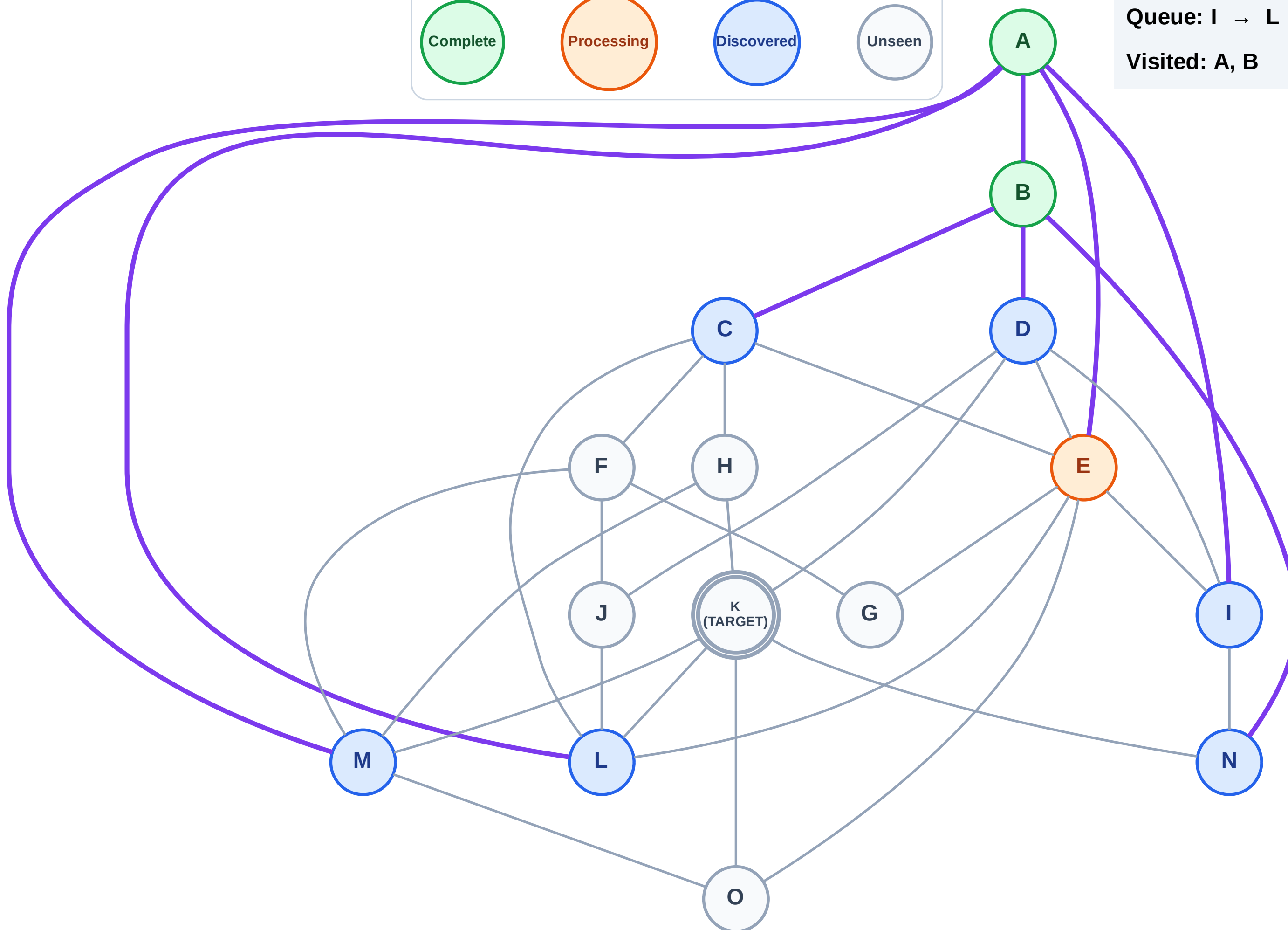
Unseen

Queue: I → L → M → C → D → N

Visited: A, B

Queue: I → L → M → C → D → N

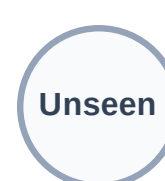
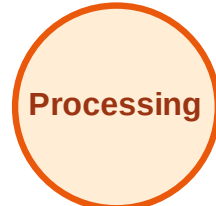
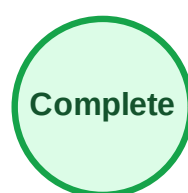
Visited: A, B



BFS Traversal

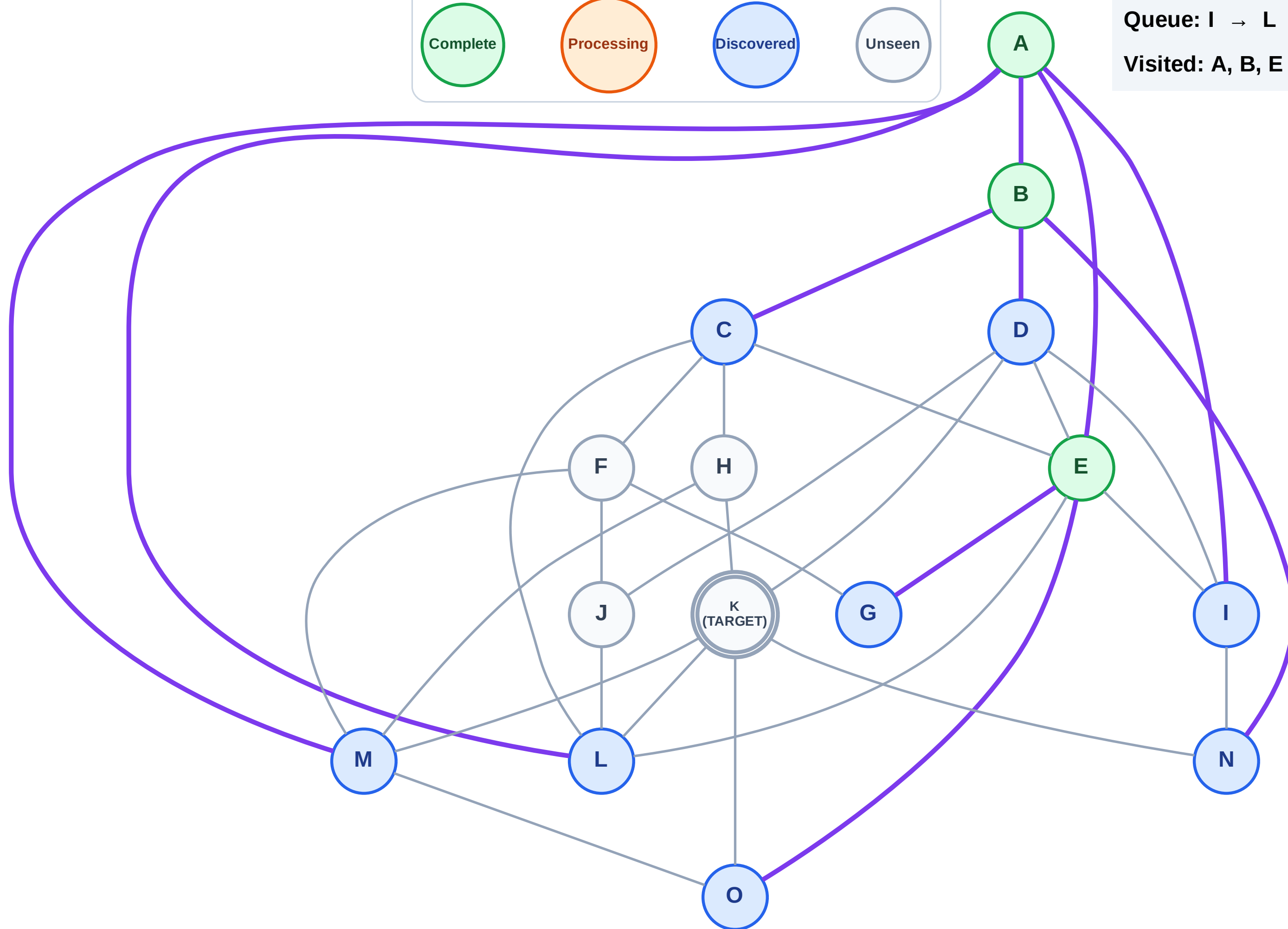
Step 7: Discover G, O from E

Legend



Queue: I → L → M → C → D → N → G → O

Visited: A, B, E



BFS Traversal

Step 8: Process node I

Legend

Complete

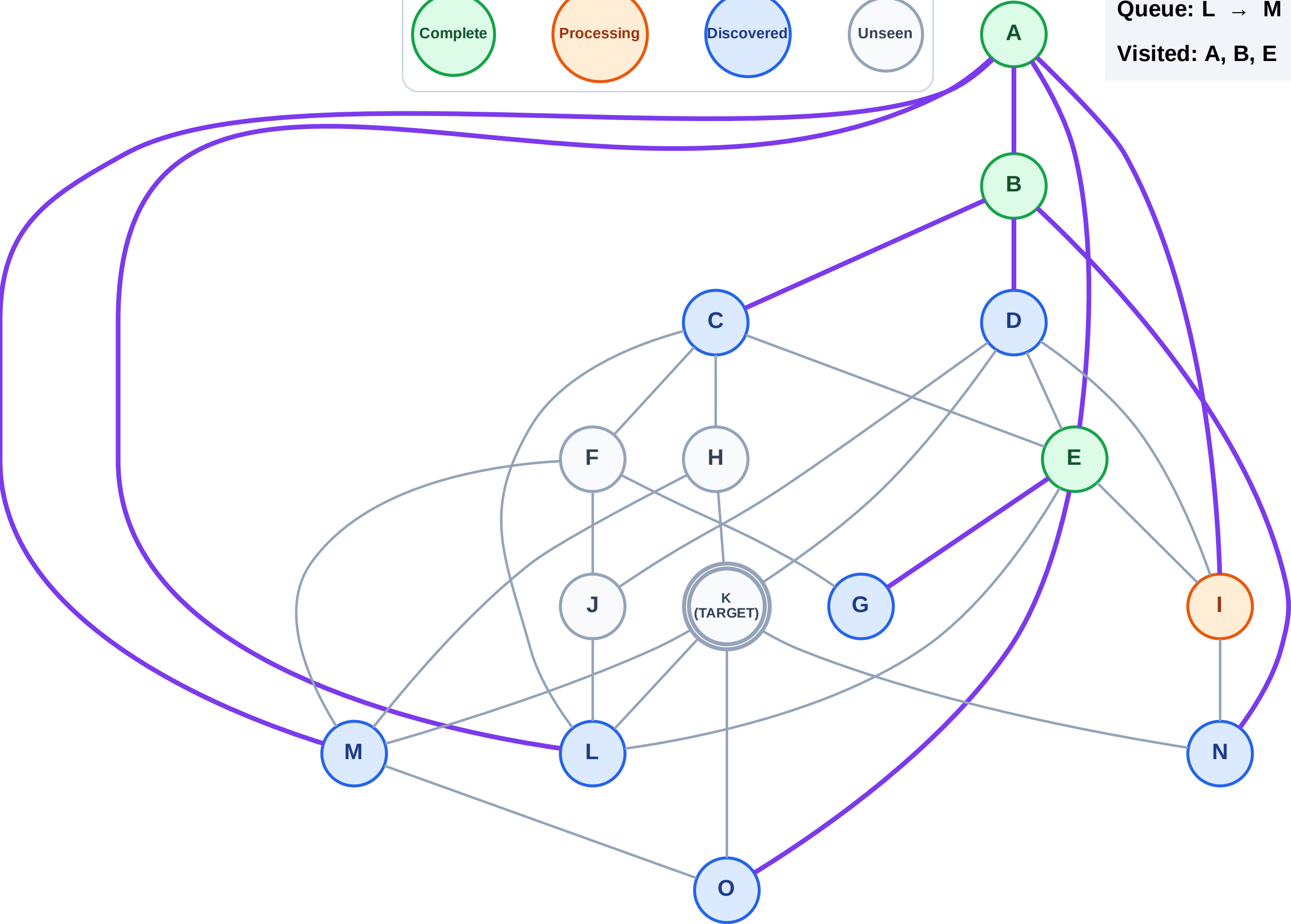
Processing

Discovered

Unseen

Queue: L → M → C → D → N → G → O

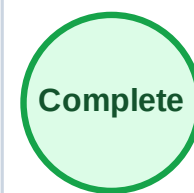
Visited: A, B, E



BFS Traversal

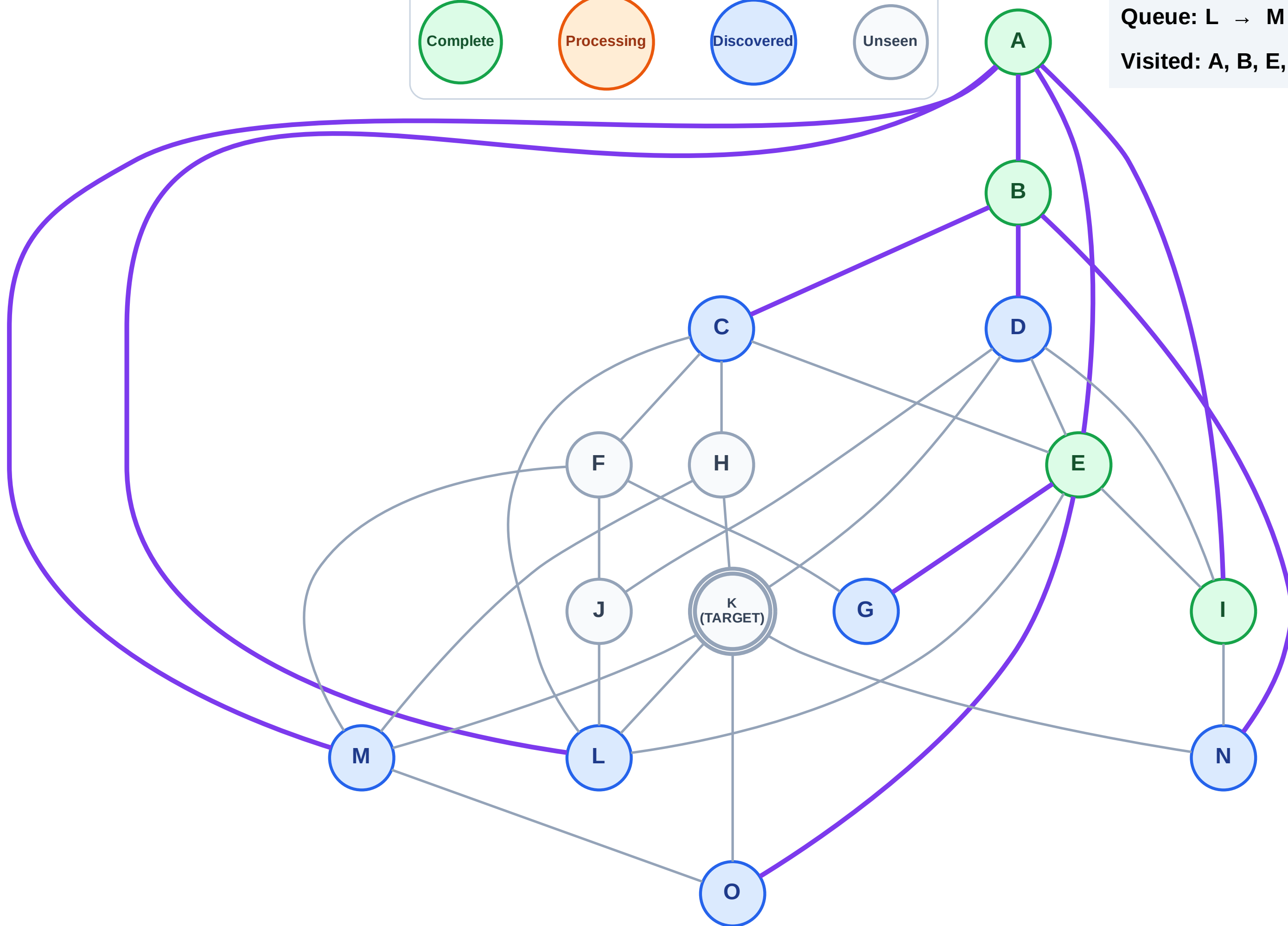
Step 9: No new neighbors from I

Legend



Queue: L → M → C → D → N → G → O

Visited: A, B, E, I



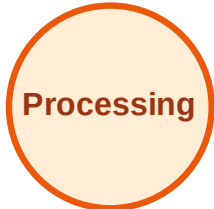
BFS Traversal

Step 10: Process node L

Legend



Complete



Processing



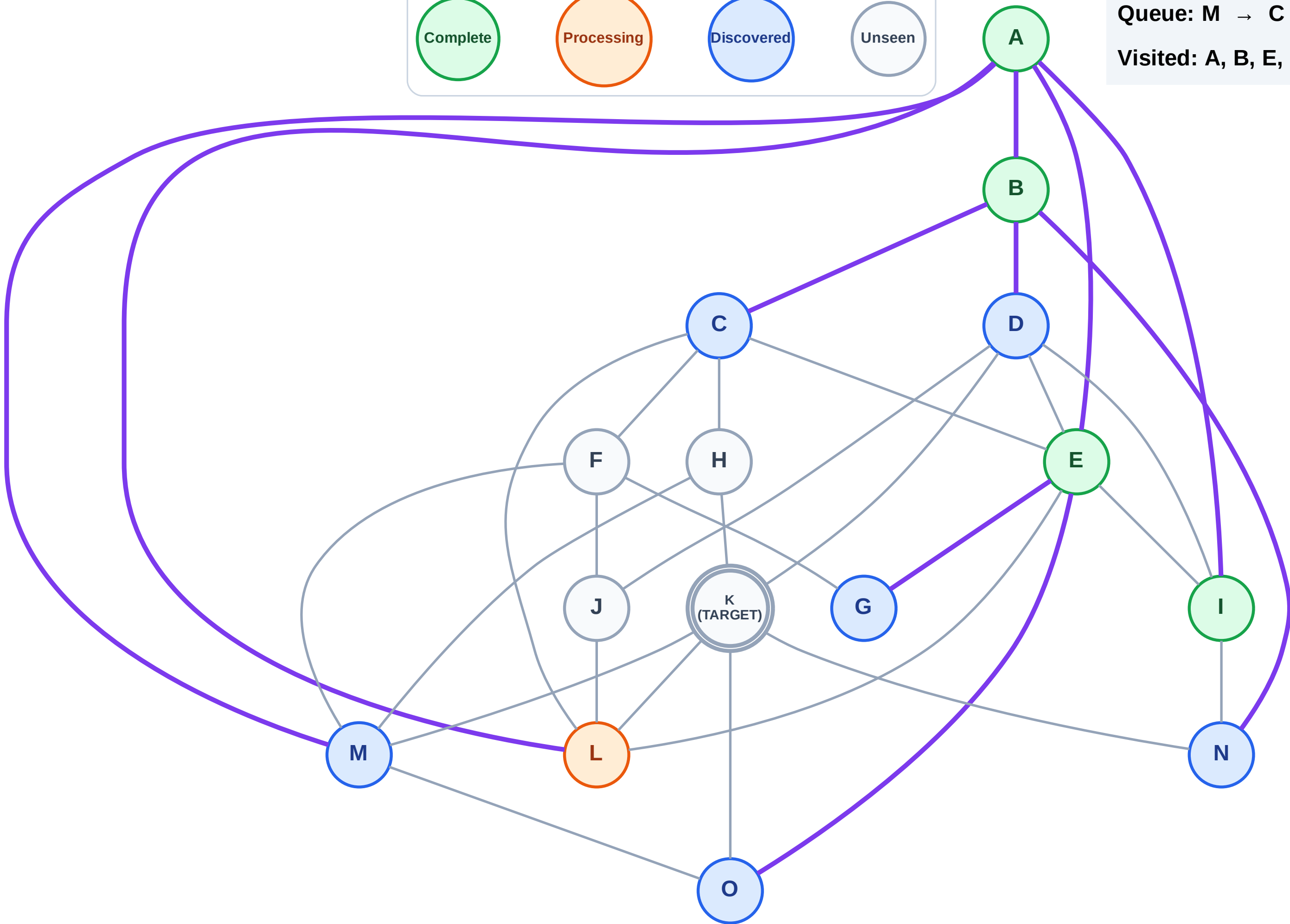
Discovered



Unseen

Queue: M → C → D → N → G → O

Visited: A, B, E, I



BFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → L → K

Legend

Complete

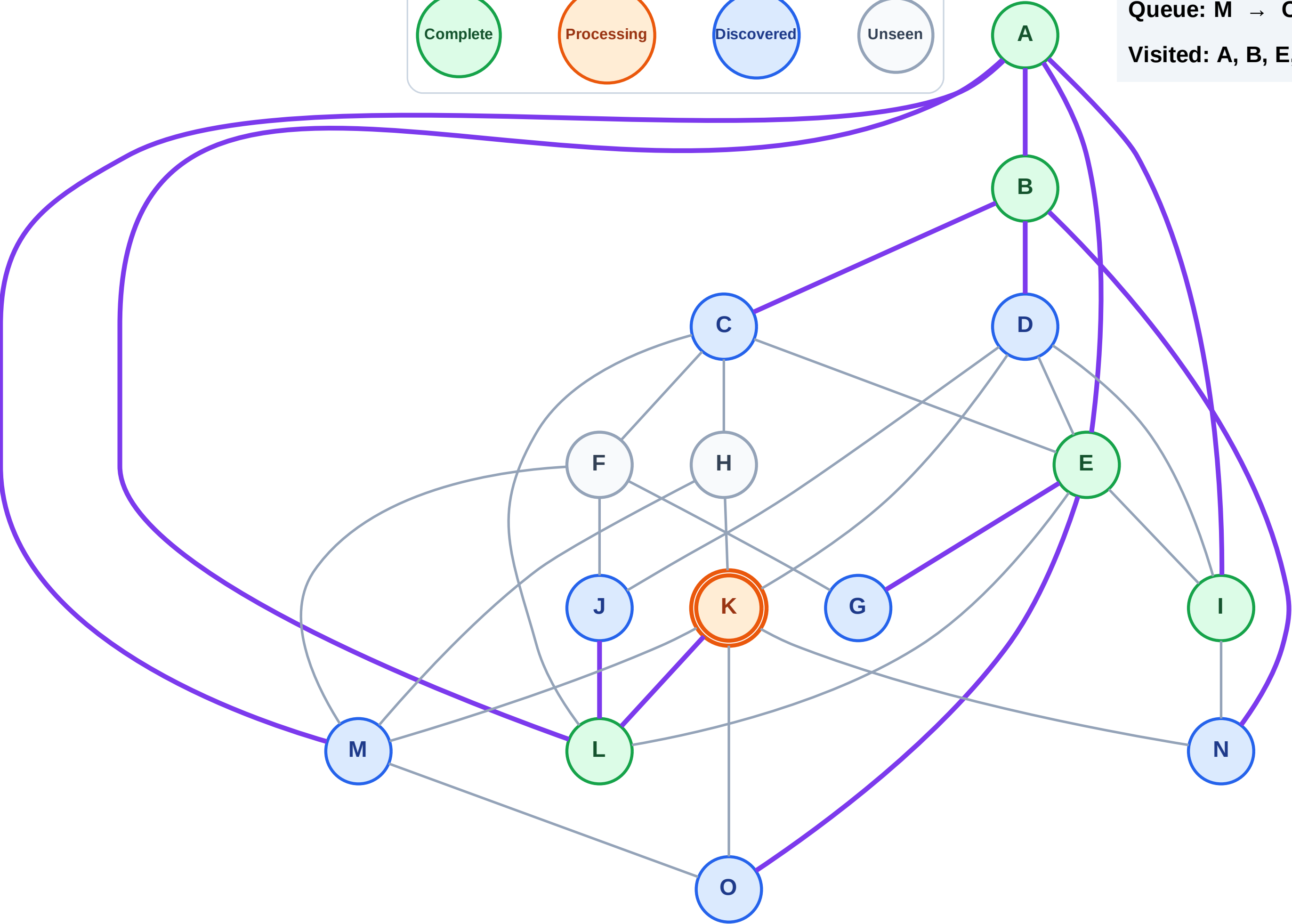
Processing

Discovered

Unseen

Queue: M → C → D → N → G → O → J

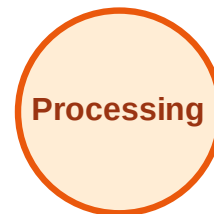
Visited: A, B, E, I, K, L



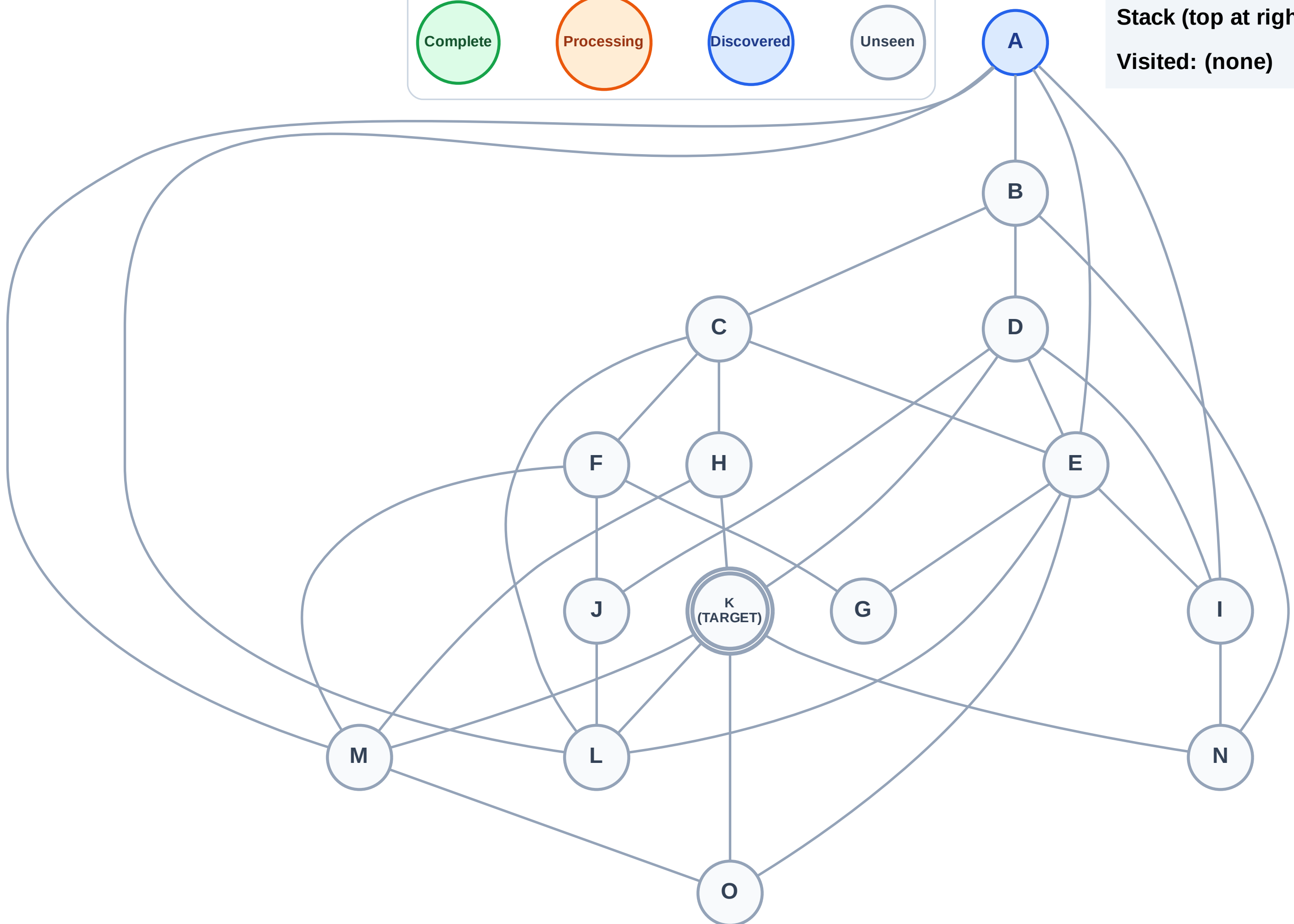
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



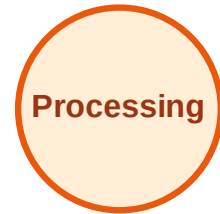
Stack (top at right): A
Visited: (none)



DFS Traversal

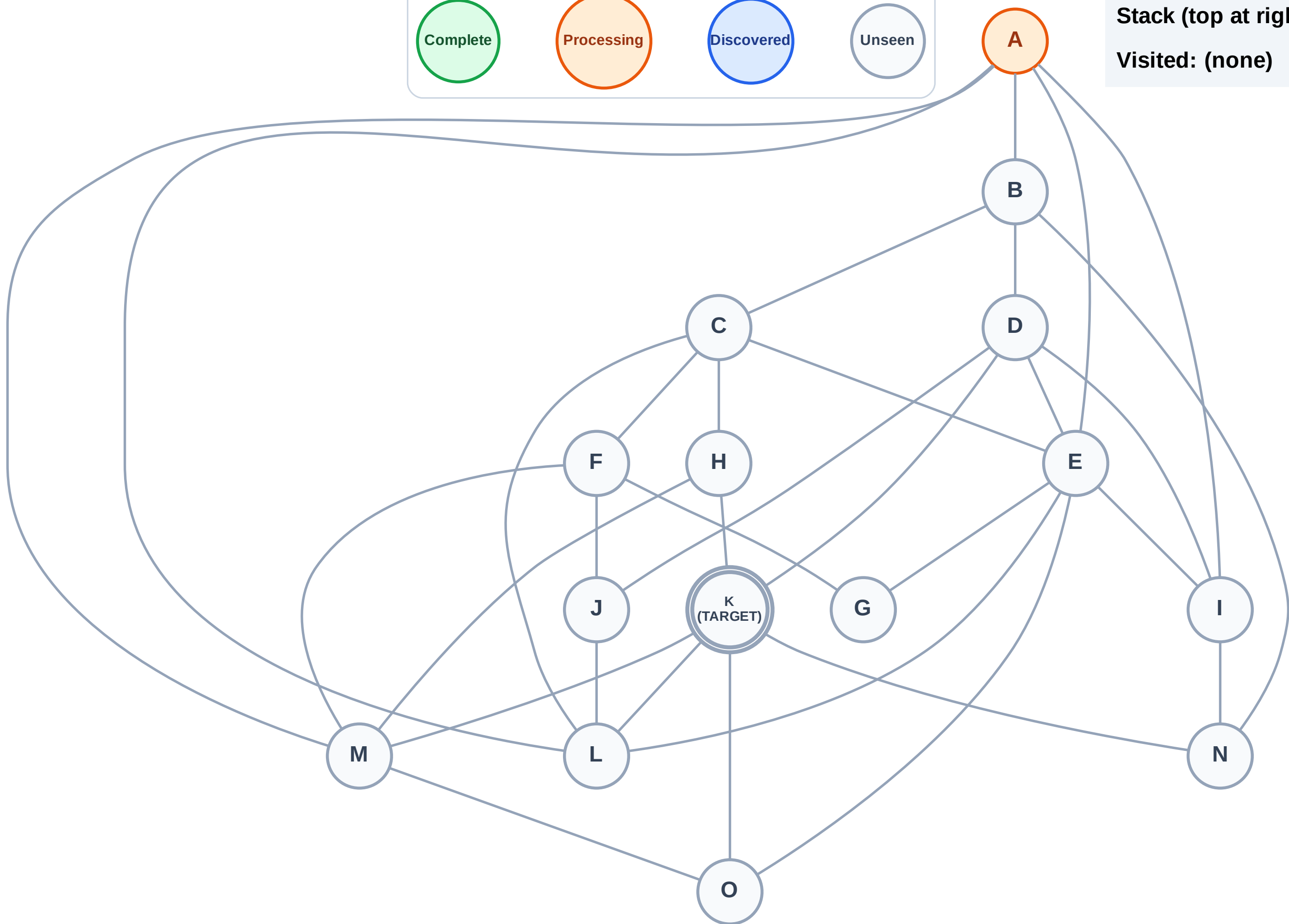
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



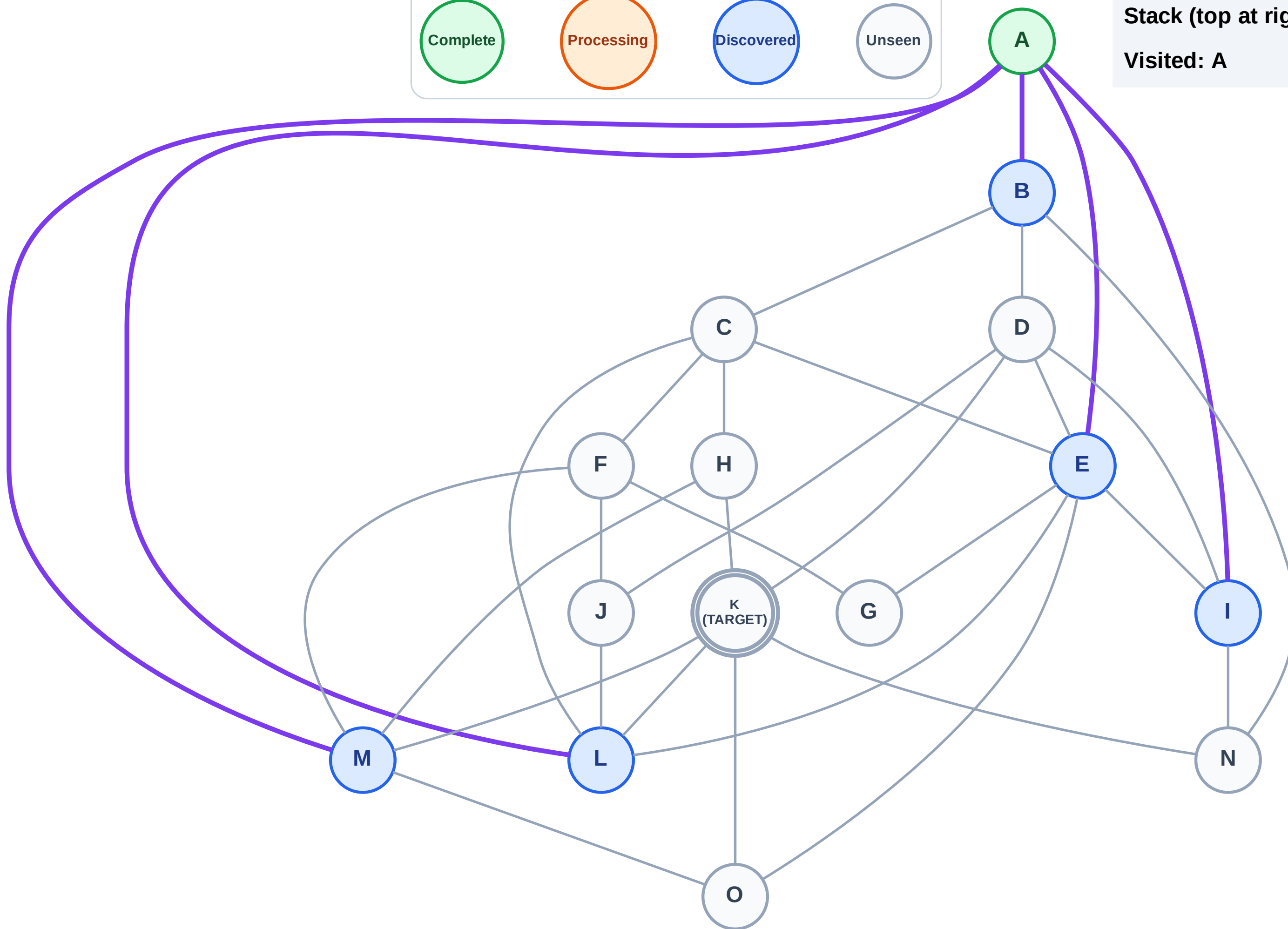
DFS Traversal

Step 3: Discover M, L, I, E, B from A

Legend



Stack (top at right): M | L | I | E | B
Visited: A



DFS Traversal

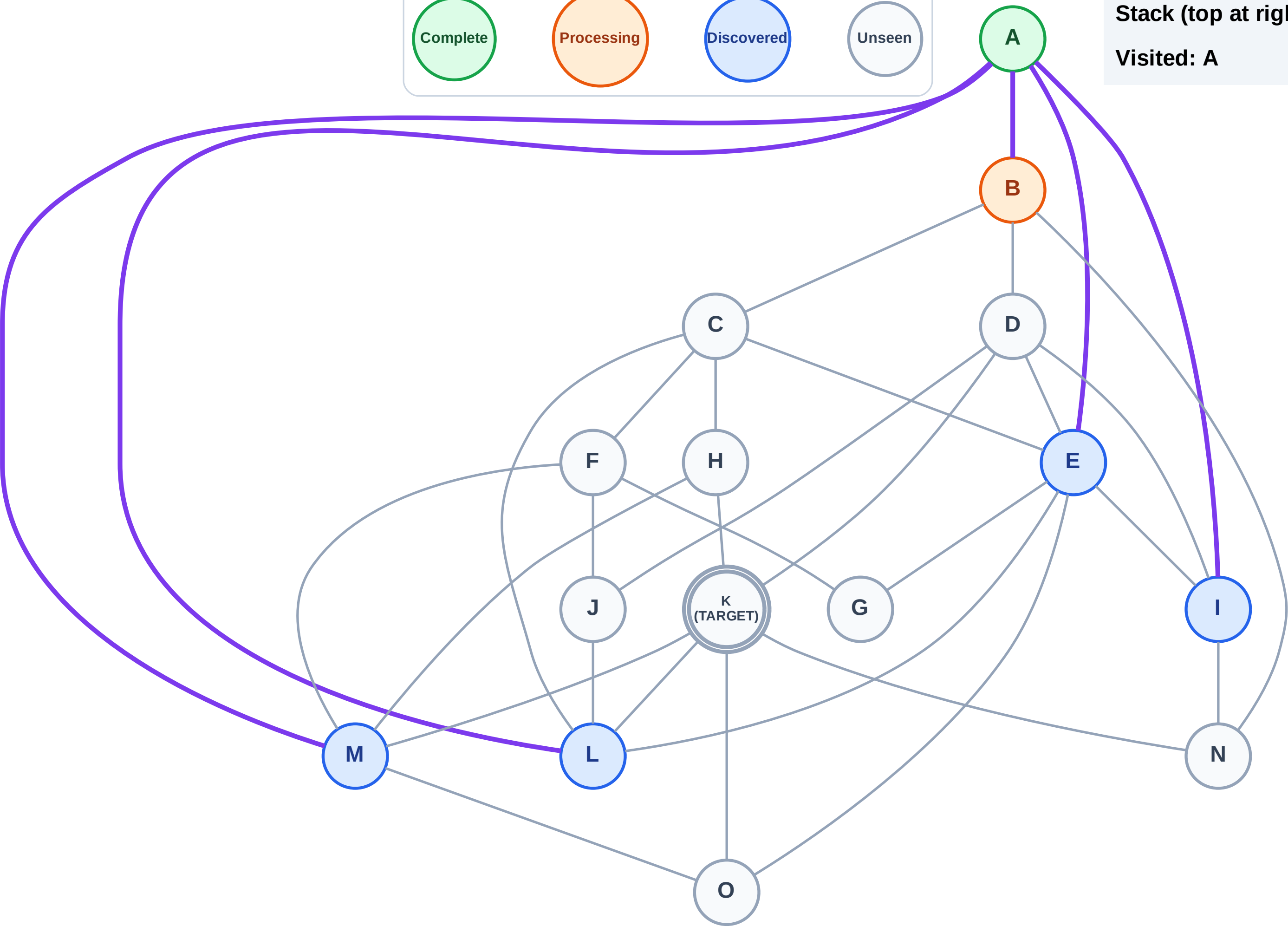
Step 4: Process node B

Legend



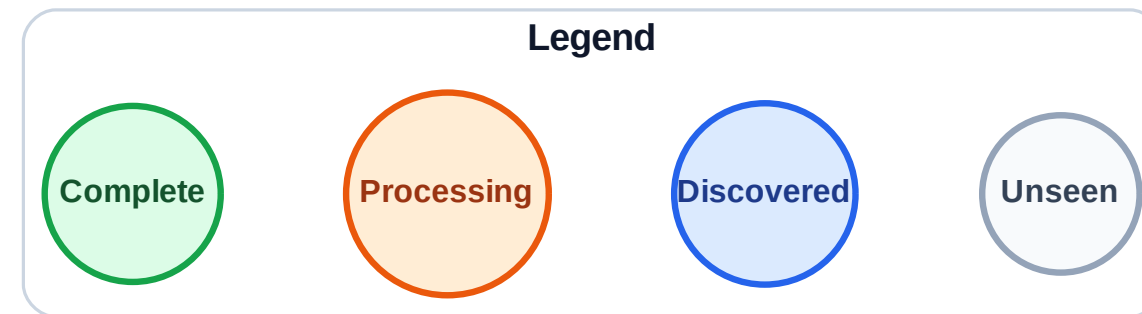
Stack (top at right): M | L | I | E

Visited: A



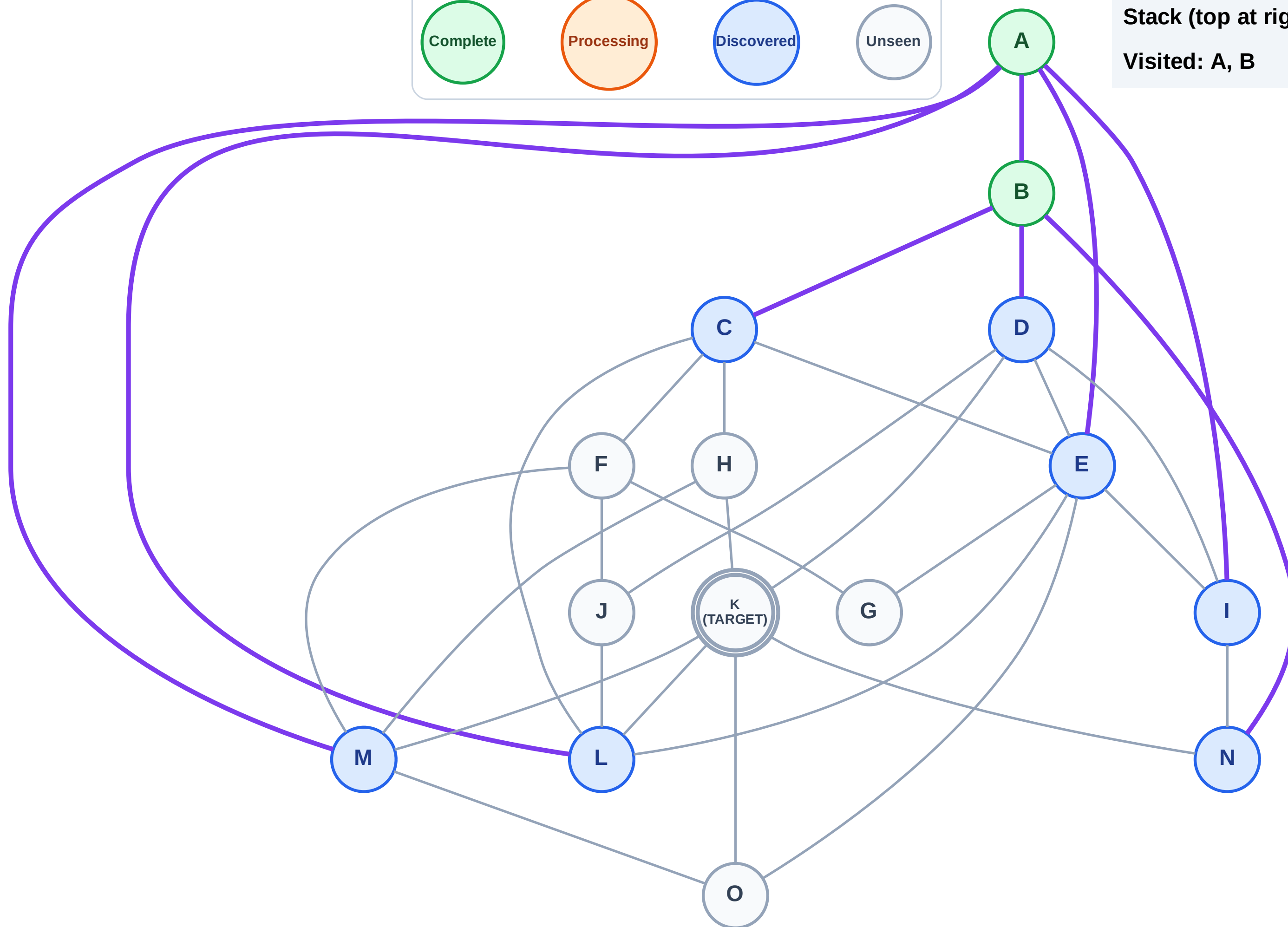
Step 5: Discover N, D, C from B

Step 5: Discover N, D, C from B



Stack (top at right): M | L | I | E | N | D | C

Visited: A, B



DFS Traversal

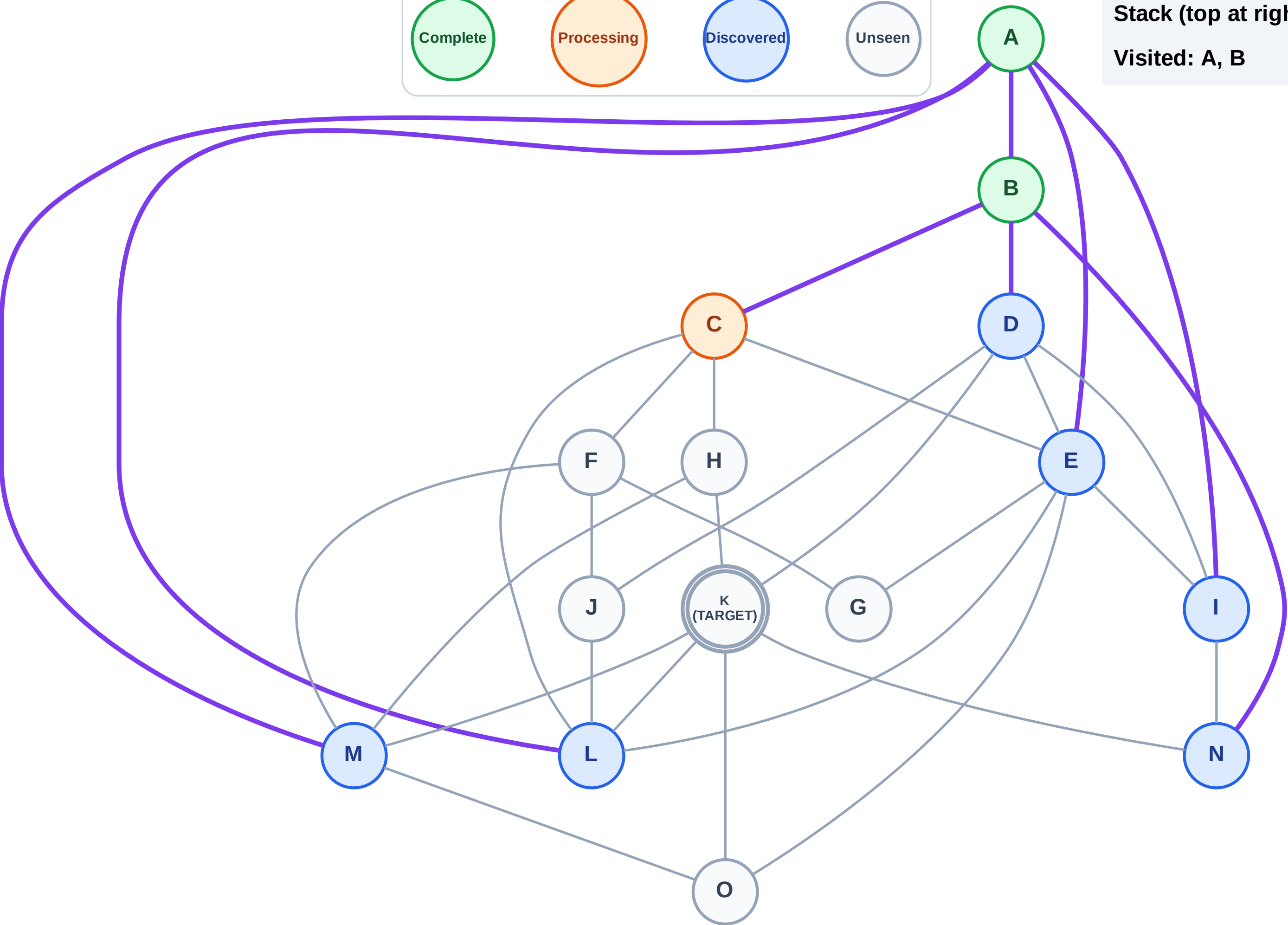
Step 6: Process node C

Legend



Stack (top at right): M | L | I | E | N | D

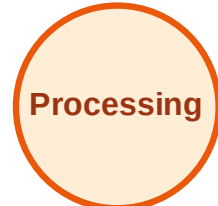
Visited: A, B



DFS Traversal

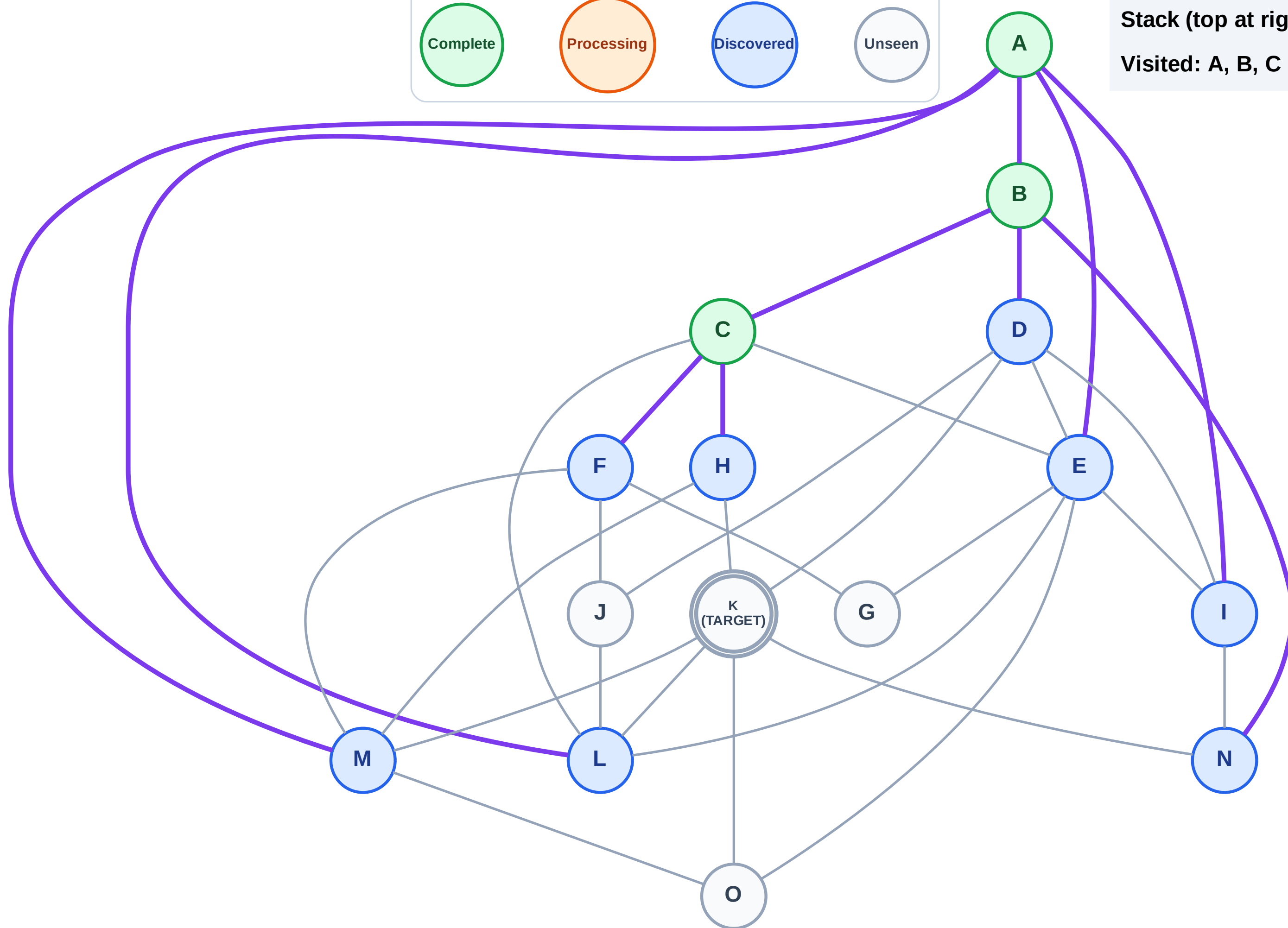
Step 7: Discover H, F from C

Legend



Stack (top at right): M | L | I | E | N | D | H | F

Visited: A, B, C



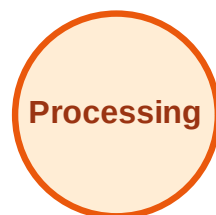
DFS Traversal

Step 8: Process node F

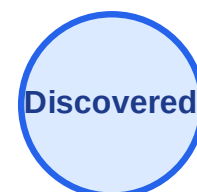
Legend



Complete



Processing



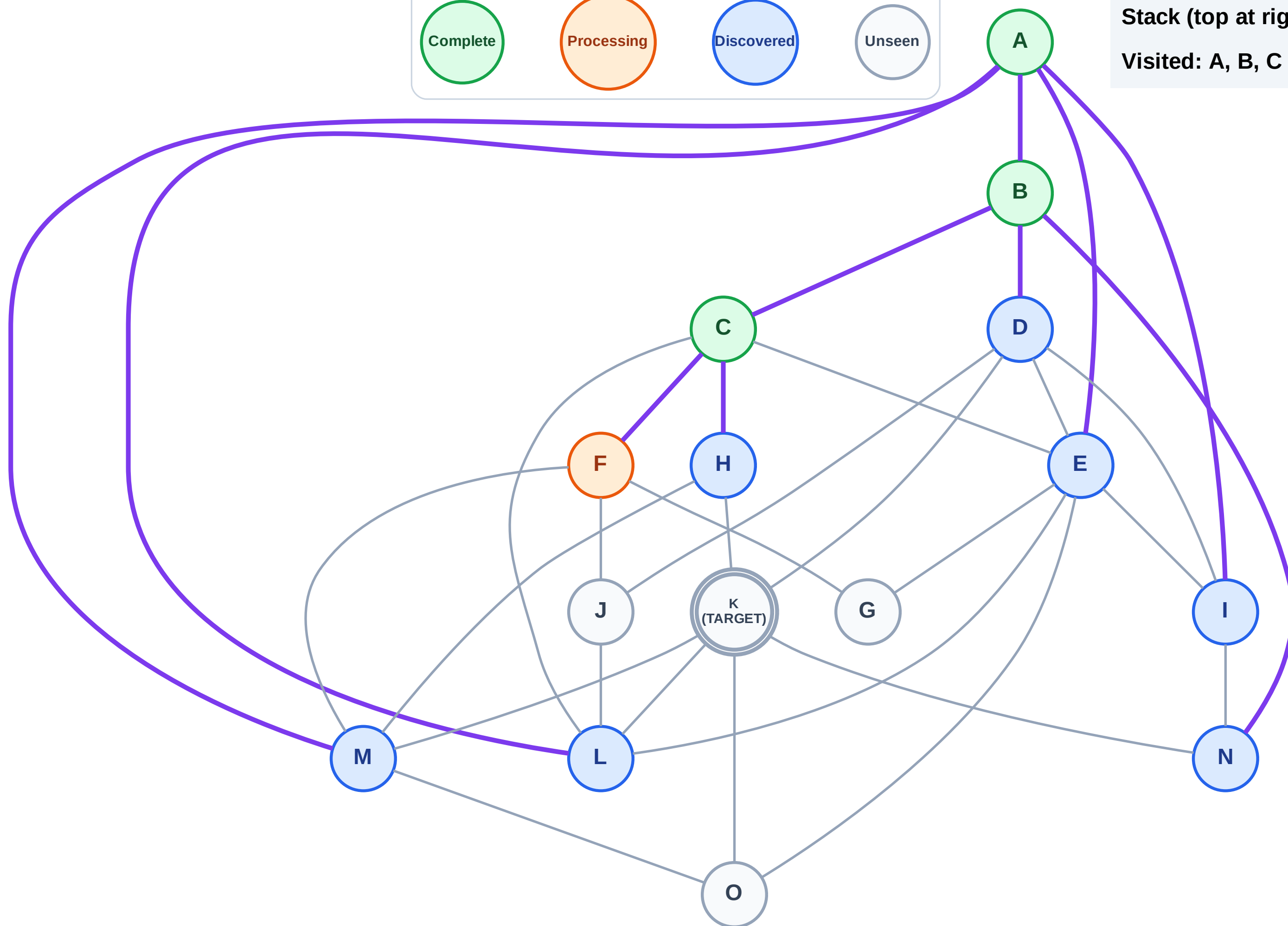
Discovered



Unseen

Stack (top at right): M | L | I | E | N | D | H

Visited: A, B, C



DFS Traversal

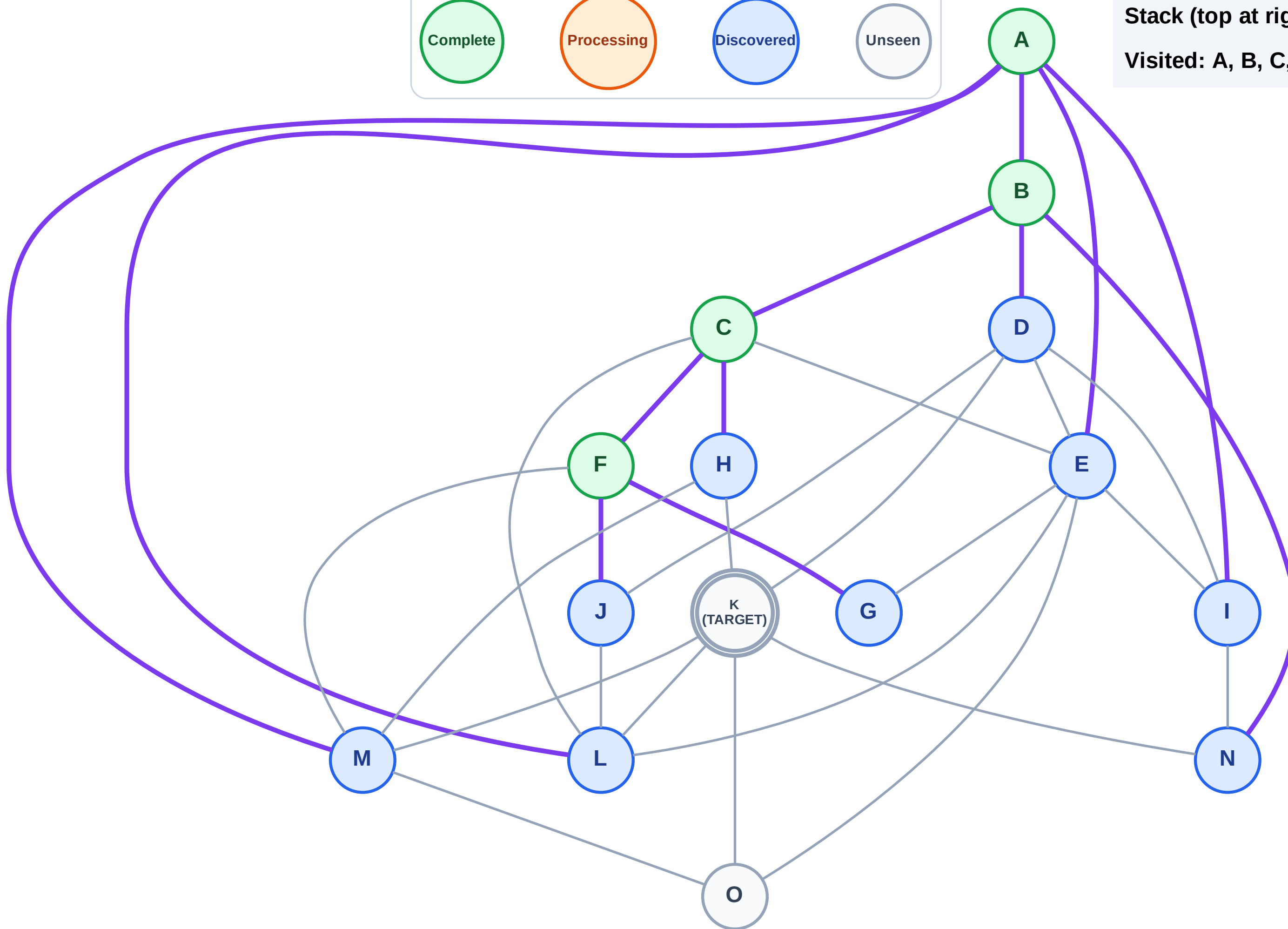
Step 9: Discover J, G from F

Legend



Stack (top at right): M | L | I | E | N | D | H | J | G

Visited: A, B, C, F



DFS Traversal

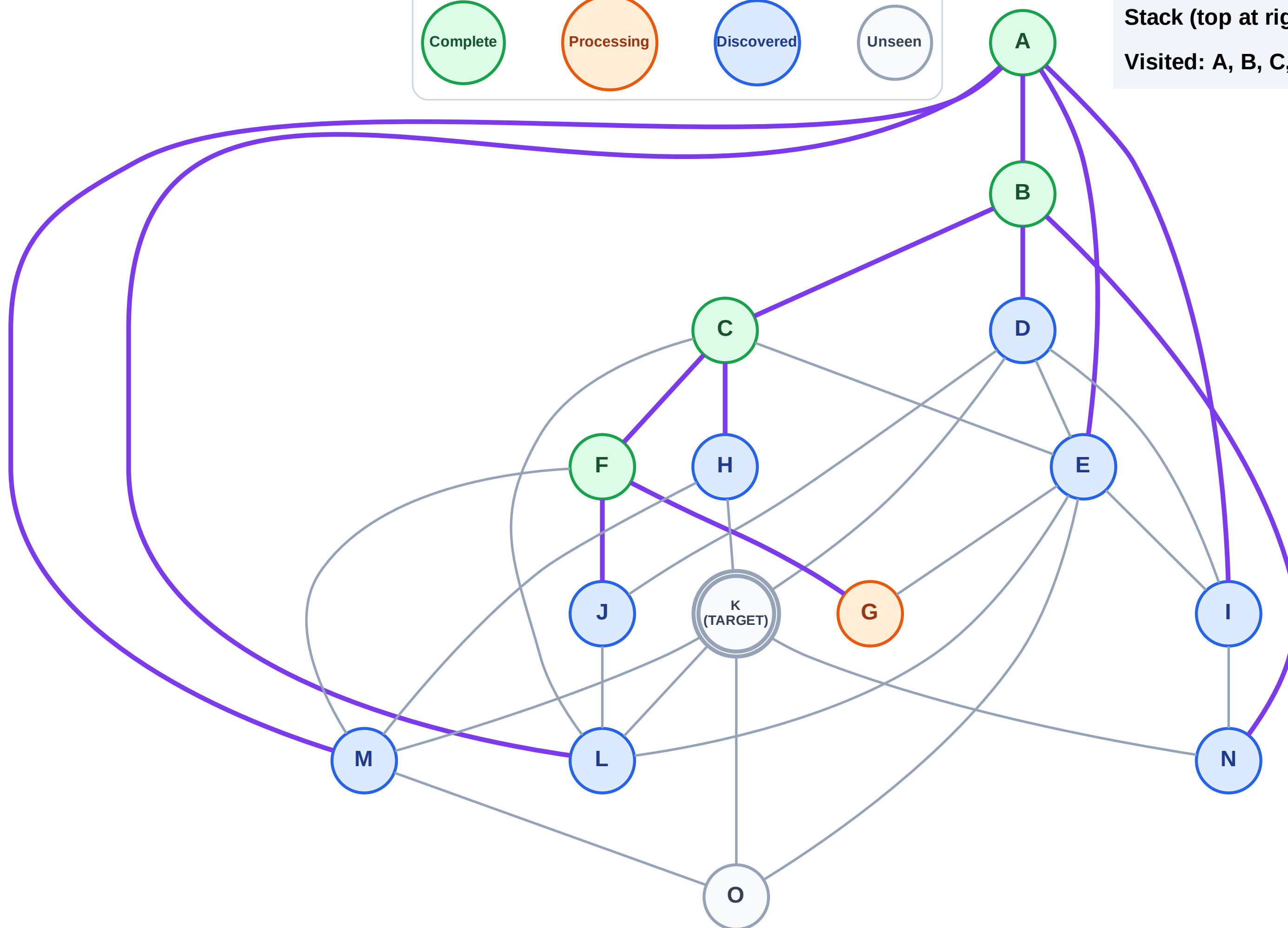
Step 10: Process node G

Legend



Stack (top at right): M | L | I | E | N | D | H | J

Visited: A, B, C, F



DFS Traversal

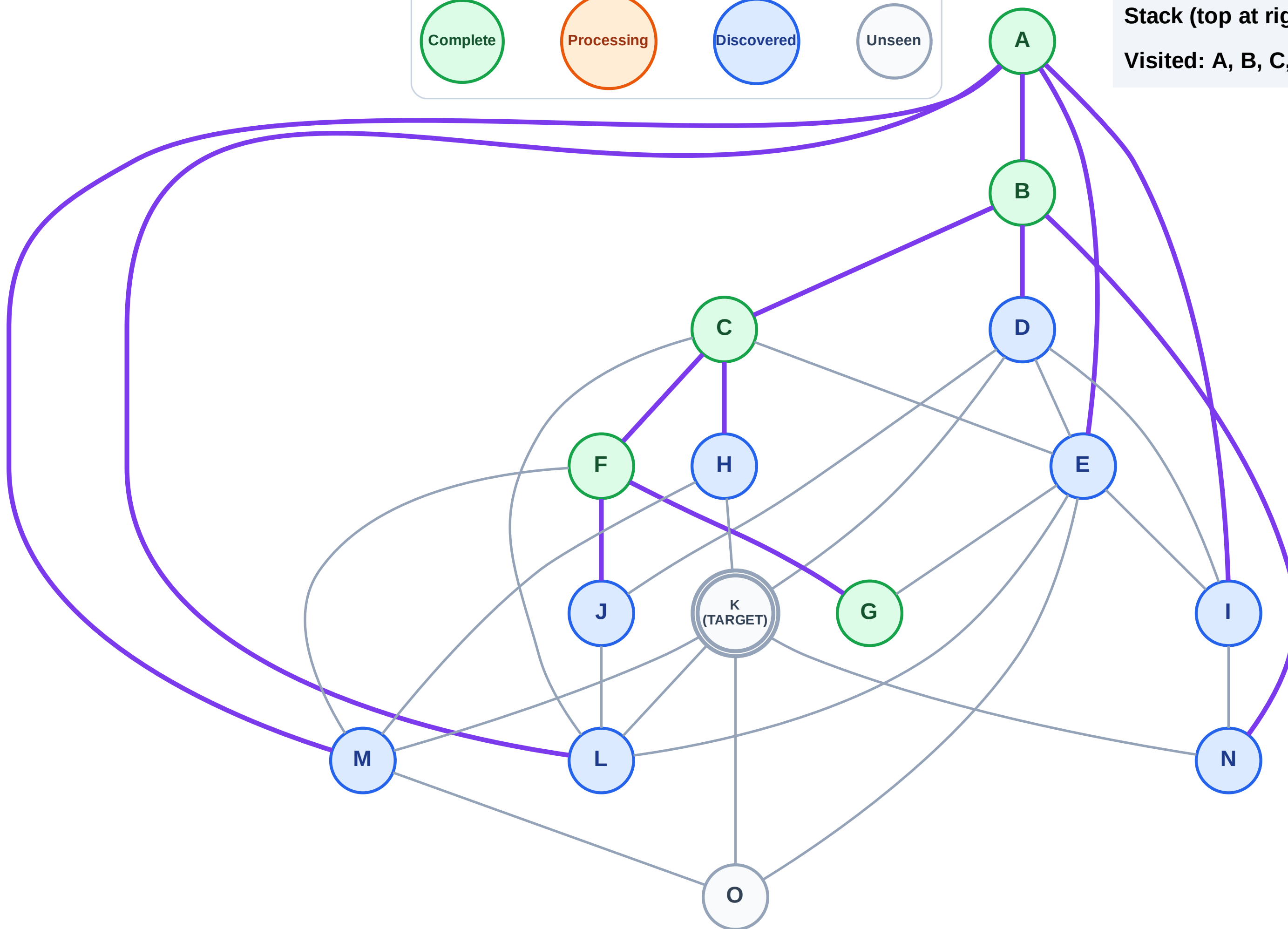
Step 11: No new neighbors from G

Legend



Stack (top at right): M | L | I | E | N | D | H | J

Visited: A, B, C, F, G



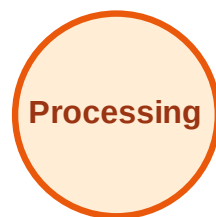
DFS Traversal

Step 12: Process node J

Legend



Complete



Processing



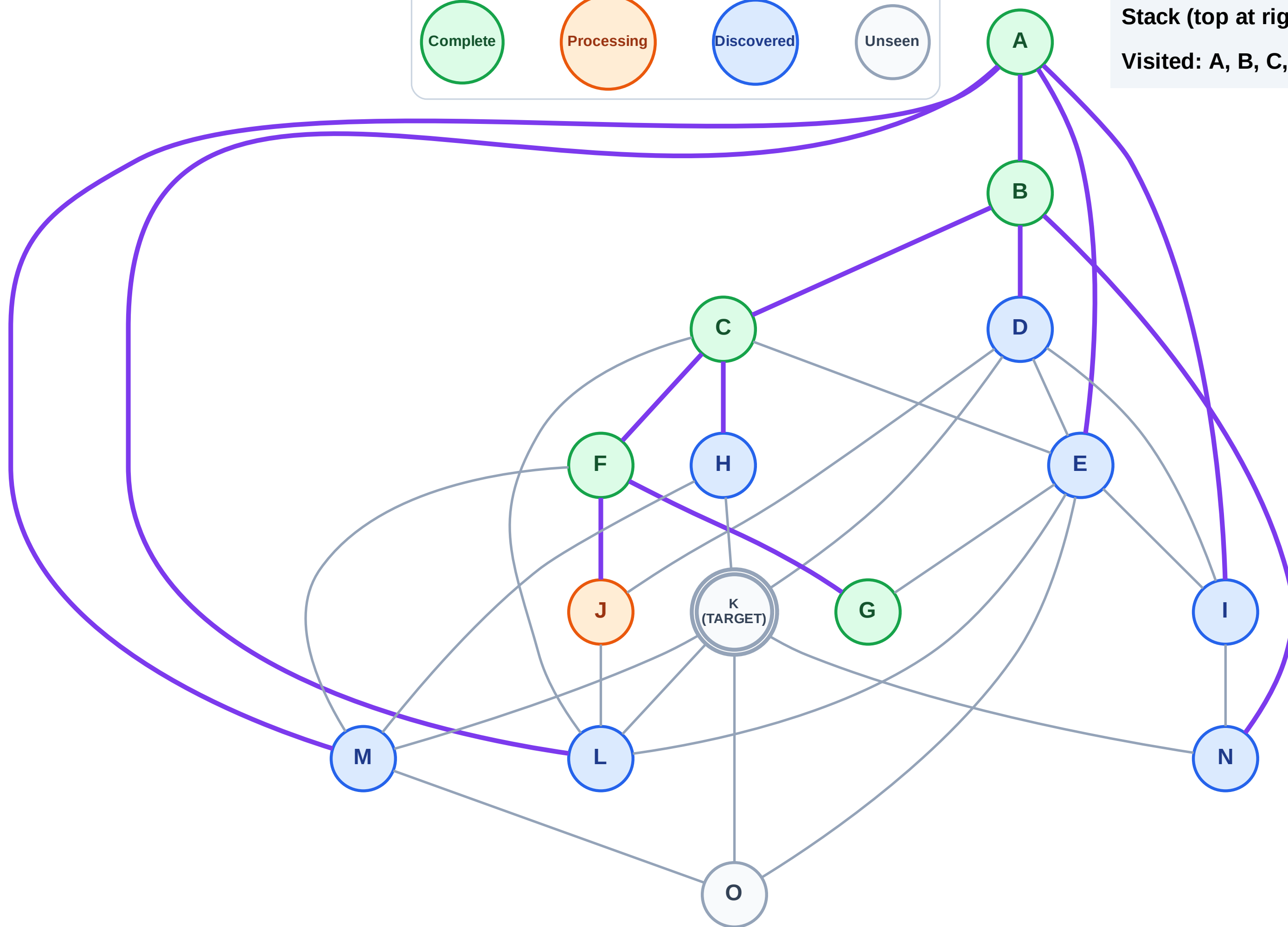
Discovered



Unseen

Stack (top at right): M | L | I | E | N | D | H

Visited: A, B, C, F, G



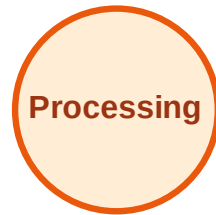
DFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



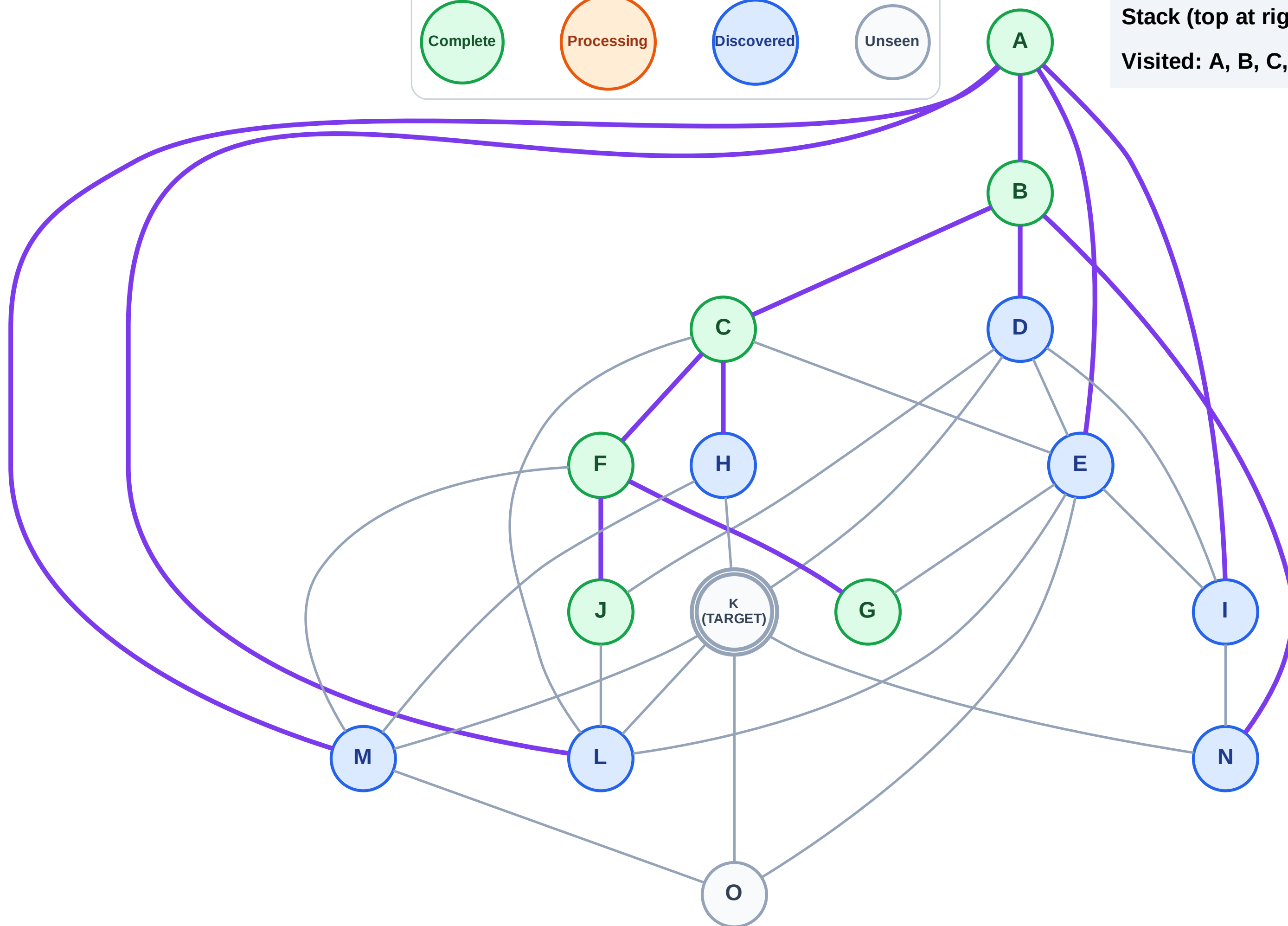
Discovered



Unseen

Stack (top at right): M | L | I | E | N | D | H

Visited: A, B, C, F, G, J



DFS Traversal

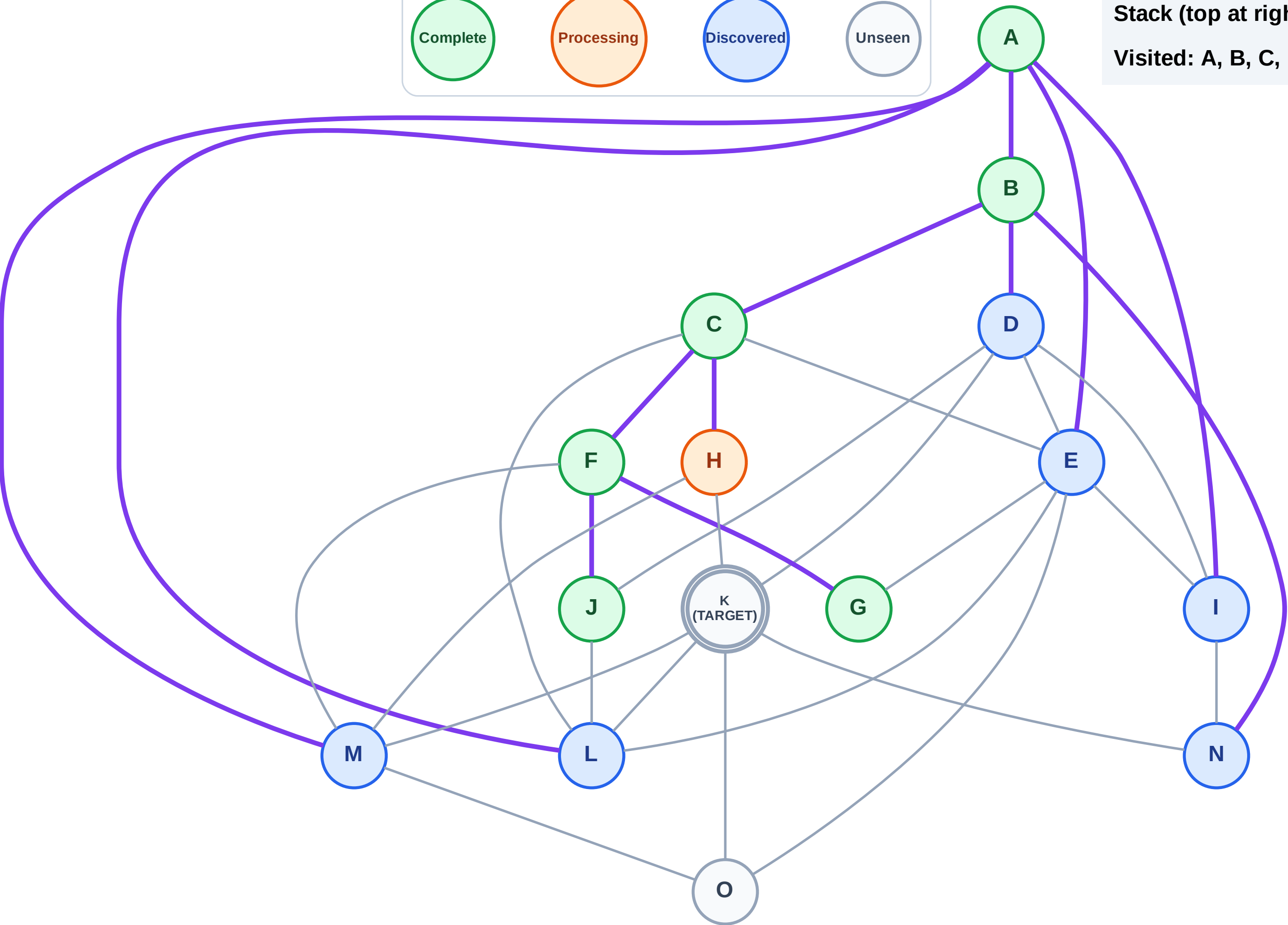
Step 14: Process node H

Legend



Stack (top at right): M | L | I | E | N | D

Visited: A, B, C, F, G, J



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → H → K

Legend



Stack (top at right): M | L | I | E | N | D

Visited: A, B, C, F, G, H, J, K

